

§0 Algebraic Foundation & Two Primitives

The foundational section of the Reconciliation Causal Framework, rewritten for v2.1. Establishes the relational algebra, the Lie–Jordan decomposition, the admissibility, the GNS construction, the algebraic fracture operator Γ , and causality/locality as structural co-primitives. Incorporates the Thm 0.4.4 rename, expanded representation-theoretic remarks, and updated acyclicity test. Five inline gap flags surface pre-existing open questions at their point of origin.

5 GAPS FLAGGED

5
SUBSECTIONS

$L \rightarrow Q$
LAYERS

7
DEFINITIONS

1
THEOREM

VERSION
AUDIENCE
DATE

v2.1 Draft (Canonical Form)
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RCF Section 0 — Algebraic Foundation and the Two Primitives

Rewritten Canonical Form — v2.1 (Draft)

Preamble — What This Section Contains and Why

This is the foundational section of the Reconciliation Causal Framework (RCF). It establishes the algebraic substrate from which all subsequent structure emerges: the relational algebra, admissibility, the GNS construction, the algebraic definition of the fracture operator F , and the naming of causality and locality as structural co-primitives grounded in the Lie–Jordan decomposition of the algebra.

What changed from v2.0

The v2.0 rewrite stripped this section to its foundational core, removing the Reconciliation Propagator R_t (§0.4 in v1.0), the Convergence Theorem (§0.5), the thin/full A_{phy} split (§0.6–0.7), the Dirac bracket (§0.7b), and the Reconciliation Principle (§0.8) — all of which depend on the fracture machinery that does not emerge until §1. v2.1 retains this stripped structure with the following corrections:

1. **Theorem 0.4.4 renamed** from "Complementarity / Principle of the Now" to "Complementarity of the Governing Mechanisms." The v2.0 name was misleading because the "now" is not defined until §4.2 (after both space §3 and time §4.1 exist). The theorem text already limited its scope to governing mechanisms; the name now matches.
2. **Remark 0.1.5 expanded** to flag the representation-theoretic question that the $D=3$ closure (§3.6) now depends on: does the Jordan algebra A_J support exactly three independent symmetric bilinear invariants? This is the hardest open question in §3 and is flagged here because the Lie–Jordan decomposition (Def 0.1.2) is its algebraic foundation.
3. **Forward-references updated** throughout to point to v2.1's revised sections: §1 (Open Expansion, Fracture, R_t — now with $F = F_{\text{intra}} + F_{\text{cross}}$ decomposition), §2 (Burden, Matter, RP — now with Petz/Bures Born-rule direction), §3 (Emergent Geometry — now with Fubini-Study metric and K_{ω^J} Jordan-built kernel).
4. **Acyclicity test updated** in §0.5 to reflect the v2.1 deferrals: the Fubini-Study metric (§3), the F decomposition (§1, §4), the adjacency structure on $\{\hat{C}_\alpha\}$ (§1), and the K_{ω^J} kernel (§3) are all deferred to their correct emergence positions.

What this section does NOT contain (deferred)

Per the v2.0 strip-and-defer policy (retained in v2.1), this section contains nothing that depends on:

- The fracture of H_{kin} ($\rightarrow$ §1.2)
- Burden as a physical quantity ($\rightarrow$ §1.3)
- The Reconciliation Propagator R_t ($\rightarrow$ §1.5)

- The reduced algebra A_{red} or M_{red} ($\rightarrow$ §1.4)
- The Dirac bracket ($\rightarrow$ §1.4.5)
- The Reconciliation Principle ($\rightarrow$ §2.1)
- The correlation kernel K_{ω} or the Fubini-Study metric ($\rightarrow$ §3)
- The adjacency structure on $\{\hat{C}_{\alpha}\}$ ($\rightarrow$ §1, from Jordan co-occurrence)

The test for this section: does it contain anything that depends on fracture, sub-Hilbert spaces, burden, R_t , A_{red} , M_{red} , the Dirac bracket, K_{ω} , or the Reconciliation Principle? If yes, the reordering has failed.

§0.0 Purpose of the Foundation

The Reconciliation Causal Framework begins with a single primitive: a relational algebra equipped with a master constraint. Physical reality is not assumed; it is derived as the constraint-compatible sector of this algebra. The foundation must therefore establish:

1. **The algebra** (what the primitive objects are and how they compose).
2. **Admissibility** (which states satisfy the constraints).
3. **The GNS construction** (how states become vectors in a Hilbert space).
4. **The fracture operator F** (algebraically — its physical interpretation is deferred).
5. **The two structural co-primitives** (causality and locality, grounded in the Lie–Jordan decomposition).

Everything else — dynamics, geometry, matter, gravity — is built on top of this foundation in later sections. The discipline of this section is to resist the temptation to introduce objects before their conceptual ingredients exist.

[GAP: v2.1 — none in §0.0. This subsection is structurally complete.]

§0.1 The Relational Algebra and Its Lie–Jordan Decomposition

Definition 0.1.1 (Relational Algebra)

A **relational algebra** is a unital complex $*$ -algebra $(A, \cdot, \dagger, 1)$ where:

- A is a complex vector space,
- $\cdot : A \times A \rightarrow A$ is an associative bilinear product,
- $\dagger : A \rightarrow A$ is an involution (antilinear, anti-automorphism),
- $1 \in A$ is the unit.

The elements of A are called **observables** (or **relations** — the two terms are used interchangeably, reflecting the framework's relational stance: an observable IS a relation between sectors of the algebra).

Definition 0.1.2 (Lie–Jordan Decomposition)

Every associative algebra A admits a canonical decomposition into two sub-algebras:

$$AB = \frac{1}{2}[A, B] + (A \circ B)$$

where:

- **The Lie bracket** (antisymmetric part): $[A, B] := AB - BA$
- **The Jordan product** (symmetric part): $A \circ B := \frac{1}{2}(AB + BA)$

The Lie sub-algebra $\mathcal{A}_L = (A, [\cdot, \cdot])$ is antisymmetric ($[A, B] = -[B, A]$) and satisfies the Jacobi identity. The Jordan sub-algebra $\mathcal{A}_J = (A, \circ)$ is symmetric ($A \circ B = B \circ A$) and satisfies the Jordan identity $(A \circ B) \circ (A \circ A) = A \circ (B \circ (A \circ A))$.

Remark 0.1.3 (Independence of Lie and Jordan structures). The Lie and Jordan structures are genuinely independent: neither is inferable from the other. This is the content of the Jordan–von Neumann–Wigner classification (1934), which shows that the Jordan algebras form a distinct algebraic species from Lie algebras, with their own representation theory. The decomposition $AB = \frac{1}{2}[A, B] + (A \circ B)$ is therefore not a trivial rewriting — it exposes two independent algebraic structures that the associative product conflates.

Remark 0.1.4 (Associativity as compatibility). Associativity of the original product $\cdot$ is the compatibility constraint between the Lie and Jordan structures: it is the condition that allows them to coexist in a single algebra. Without associativity, the Lie and Jordan products would not necessarily compose consistently. The framework treats associativity as a structural given — it is the "glue" that makes the two-primitive picture possible.

Remark 0.1.5 (The representation-theoretic question — FLAGGED). The Lie–Jordan decomposition is the algebraic foundation for the $D=3$ closure argument in §3.6. The proposed v2.1 fix for $D=3$ is that the Gram matrix of the Jordan-built kernel K_{ω}^J (Definition 3.1.2, v2.1) has rank ≤ 3 because the Jordan algebra $\mathcal{A}_J$ supports exactly three independent symmetric bilinear invariants. [GAP: v2.1 — this representation-theoretic claim ("A_J supports exactly 3 independent symmetric bilinear invariants") is not yet proven. It is the hardest open question in the $D=3$ closure argument. The Jordan–von Neumann–Wigner classification gives the structure theory for finite-dimensional Jordan algebras, but the framework's A may be infinite-dimensional, and the invariant-counting argument needs development. See §3.6 Remark 3.6.5 and Phase C Program Ch 4.]

Definition 0.1.6 (Primitive Constraints and the Master Constraint)

A **primitive constraint family** is a set $\{\hat{C}_{\alpha}\}_{\alpha \in I}$ of elements of A (indexed by a set I) such that each $\hat{C}_{\alpha}$ represents a relational consistency condition. The **master constraint** is the collective:

$$\hat{M} := \sum_{\alpha \in I} w_{\alpha} \cdot \hat{C}_{\alpha}^{\dagger} \hat{C}_{\alpha}$$

where $\{w_{\alpha}\}$ are positive weights. The master constraint is a positive operator: $\hat{M} \geq 0$.

Remark 0.1.7 (M is a collective, not a single object). The master constraint is not a single irreducible object; it is a **collective** — a weighted sum over the sub-constraint ideals generated by each $\hat{C}_{\alpha}$. The Tier 1 / Tier 2 separation of $\hat{M}$ into its sub-collectives (the part that gets quotiented away by the Dirac bracket and the part that survives on A_{red}) is deferred to §1.4, where

the fracture machinery makes this separation operationally meaningful.

[GAP: v2.1 — the adjacency structure on the constraint index set I is not yet defined. This is the shared prerequisite for T-1 (Lieb-Robinson bounds, §1) and T-2 (detectability lemma, §1.6). The proposed source: Jordan co-occurrence — two constraints $\hat{C}_\alpha, \hat{C}_\beta$ are "adjacent" if $\hat{C}_\alpha \circ \hat{C}_\beta \neq 0$. This adjacency structure is defined in §1.1 (after the Open Expansion Principle activates the Lie–Jordan decomposition). See Phase C Program Ch 6–7.]

§0.2 Admissibility and the Constraint Ideal

Definition 0.2.1 (Algebraic State)

An **algebraic state** on A is a positive linear functional $\omega: A \rightarrow \mathbb{C}$ with $\omega(1) = 1$. The set of all algebraic states is denoted $S(A)$.

Definition 0.2.2 (Admissibility — Zero-Constraint Compatibility)

A state $\omega \in S(A)$ is **admissible** (or **physical**) if it satisfies the Master-Zero condition:

$$\omega(\hat{M}) = 0$$

Equivalently (since $\hat{M} = \sum_\alpha w_\alpha \hat{C}_\alpha^\dagger \hat{C}_\alpha \geq 0$):

$$\omega(\hat{C}_\alpha^\dagger \hat{C}_\alpha) = 0 \quad \text{for all } \alpha \in I$$

The set of admissible states is denoted $S_{\text{adm}}(A) \subset S(A)$.

Remark 0.2.3 (Kinematic vs. admissible states). Not every algebraic state is admissible. A **kinematic state** is any $\omega \in S(A)$; it need not satisfy $\omega(\hat{M}) = 0$. The kinematic state space $S(A)$ is the "unconstrained" space; the admissible state space $S_{\text{adm}}(A)$ is the "physical" target. The dynamics that connects them (the Reconciliation Propagator R_t , which drives kinematic states toward admissible ones) is constructed in §1.5.

Definition 0.2.4 (Constraint Ideal)

The **constraint ideal** is the left ideal:

$$\mathcal{I}_C := \{A \in A : \omega(A^\dagger A) = 0 \text{ for all } \omega \in S_{\text{adm}}(A)\}$$

This is the set of elements that vanish in every admissible state — the "unphysical" directions of A .

Definition 0.2.5 (Compatibility Bracket)

The **compatibility bracket** between two constraints $\hat{C}_\alpha, \hat{C}_\beta$ is:

$$\Delta_{\alpha\beta} := [\hat{C}_\alpha, \hat{C}_\beta] = \hat{C}_\alpha \hat{C}_\beta - \hat{C}_\beta \hat{C}_\alpha$$

This is a Lie bracket (antisymmetric, satisfies Jacobi). It measures the **Lie-algebraic obstruction** between two constraint sectors: if $\Delta_{\alpha\beta} \neq 0$, the constraints cannot be simultaneously satisfied in a single representation without residual structure.

Remark 0.2.6 (Tier 1 / Tier 2 distinction — deferred). When the compatibility matrix $\Delta^{(1)}_{\alpha\beta} = [\hat{C}_\alpha, \hat{C}_\beta]$ is invertible on a subset of constraints, those constraints are second-class (Tier 1) and will be quotiented away by the Dirac bracket (§1.4.5). The remaining constraints are first-class (Tier 2) and survive on the

reduced algebra A_{red} (§1.4.2). This Tier 1/Tier 2 distinction is not made at Layer L because it requires the fracture machinery of §1 to be operationally meaningful. [See §1.4 for the formal definition.]

Remark 0.2.7 (Jordan component of the constraint algebra). The constraint algebra inherits both Lie and Jordan components from the Lie–Jordan decomposition of A. The Lie component $\Delta_{\alpha\beta} = [\hat{C}_{\alpha}, \hat{C}_{\beta}]$ governs the obstruction structure (and appears in F, Definition 0.3.6). The Jordan component $\hat{C}_{\alpha} \circ \hat{C}_{\beta}$ governs the sectoral decomposition — it determines which constraints are "co-occurrent" (adjacent in the emergent geometry) and which are disjoint. This Jordan component becomes the adjacency structure on the constraint index set I that is needed for T-1 (Lieb–Robinson) and T-2 (detectability lemma). [Defined formally in §1.1 after the Open Expansion Principle activates the decomposition.]

[GAP: v2.1 — the Jordan component $\hat{C}_{\alpha} \circ \hat{C}_{\beta}$ is named here but not yet given operational meaning. The formal definition of the adjacency graph on I (from Jordan co-occurrence) is deferred to §1.1. This is the shared prerequisite for T-1 and T-2. If the adjacency structure cannot be made rigorous, both T-1 and T-2 stall. See Phase C Program Ch 6–7.]

§0.3 GNS Construction and the Emergent Hilbert Space

Definition 0.3.1 (GNS Representation)

Given an algebraic state $\omega \in S(A)$, the **Gel'fand–Naimark–Segal (GNS) construction** produces a Hilbert space $\mathcal{H}_{\omega}$, a $*$ -representation $\pi_{\omega}: A \rightarrow \mathcal{B}(\mathcal{H}_{\omega})$, and a cyclic vector $\Omega_{\omega} \in \mathcal{H}_{\omega}$ such that:

$$\omega(A) = \langle \Omega_{\omega} | \pi_{\omega}(A) | \Omega_{\omega} \rangle$$

The GNS Hilbert space is constructed as the completion of $A / \mathcal{N}_{\omega}$ where $\mathcal{N}_{\omega} = \{A \in A : \omega(A^{\dagger}A) = 0\}$ is the null ideal, equipped with the inner product:

$$\langle [A]_{\omega} | [B]_{\omega} \rangle_{\omega} := \omega(A^{\dagger}B)$$

Definition 0.3.2 (Kinematic Hilbert Space)

The **kinematic Hilbert space** is the GNS Hilbert space $\mathcal{H}_{\text{kin}} := \mathcal{H}_{\omega_{\text{kin}}}$ constructed from a chosen kinematic state $\omega_{\text{kin}} \in S(A)$. The kinematic state is not required to be admissible — it is the "unconstrained" starting point from which the Reconciliation Propagator R_t (§1.5) drives the system toward the admissible sector.

Definition 0.3.3 (GNS Inner Product and the Correlation Strength)

The **GNS inner product** on $\mathcal{H}_{\text{kin}}$ is:

$$\langle A, B \rangle_{\omega_{\text{kin}}} := \omega_{\text{kin}}(A^{\dagger}B)$$

The **correlation strength** between two observables is:

$$s(A, B) := \frac{|\omega_{\text{kin}}(A^{\dagger}B)|^2}{\omega_{\text{kin}}(A^{\dagger}A) \cdot \omega_{\text{kin}}(B B^{\dagger})}$$

This is the squared cosine of the angle between $[A]_{\omega}$ and $[B]_{\omega}$ in the GNS Hilbert space. It is an algebraic quantity — available at Layer Q — but its full geometric interpretation (as the correlation kernel K_{ω} from which the emergent distance is derived) is deferred to §3.

Remark 0.3.4 (K_ω vs. K_ω^J — the Jordan-built question). The v2.0 correlation kernel $K_\omega(A, B) = s(A, B) = |\omega_{\text{kin}}(A^\dagger B)|^2 / (\cdots)$ uses the raw sesquilinear form $\omega_{\text{kin}}(A^\dagger B)$ — it does not use the Jordan product $A \circ B$ anywhere. The v2.0 Remark 3.1.3 claimed the symmetry of K_ω "comes from the Jordan structure," but the proof it gave was pure state-Hermiticity ($|\omega(A^\dagger B)| = |\omega(B^\dagger A)|$), which never touches $A \circ B$.

v2.1 proposes the Jordan-built variant (Definition 3.1.2, v2.1):

$$K_\omega^J(A, B) := \frac{\omega_{\text{kin}}(A \circ B)}{\sqrt{\omega_{\text{kin}}(A^\dagger A) \cdot \omega_{\text{kin}}(B^\dagger B)}}$$

This genuinely uses the Jordan product $A \circ B = \frac{1}{2}(AB + BA)$. It is symmetric by construction ($A \circ B = B \circ A$). The Fubini-Study distance $d_{\text{FS}} = \ell_0 \cdot \arccos(|K_\omega^J|)$ is still a metric (arccos on any inner-product space). The $D=3$ Gram-rank argument then becomes a representation-theoretic question about Jordan algebra invariants (see Remark 0.1.5).

[GAP: v2.1 — whether K_ω^J is the correct kernel for the framework's geometry is not yet determined. The question is: does K_ω^J (Jordan-built) or K_ω (sesquilinear) better captures the "where"-structure (locality)? The framework's two-primitive picture says locality = Jordan, so K_ω^J is the principled choice. But the downstream formulas (cubic kernel, distance, triangle inequality) need to be re-derived with K_ω^J , and this work is done in §3. See §3.1 Remark 3.1.3 (v2.1) for the full discussion.]

Definition 0.3.5 (The Fracture Operator — Algebraic)

The **fracture operator** is the positive operator on $\mathcal{H}_{\text{kin}}$:

$$\hat{F} := \sum_{\alpha, \beta} \pi_{\text{kin}}(\Delta_{\alpha\beta})^\dagger \pi_{\text{kin}}(\Delta_{\alpha\beta})$$

where $\Delta_{\alpha\beta} = [\hat{C}_\alpha, \hat{C}_\beta]$ is the compatibility bracket (Definition 0.2.5). Since each term is positive ($\pi(\Delta)^\dagger \pi(\Delta) \geq 0$), we have $\hat{F} \geq 0$.

Remark 0.3.6 (Algebraic availability vs. physical interpretation). $\hat{F}$ is **algebraically available** at Layer Q: the definition requires only the GNS representation (§0.3.1) and the compatibility brackets (§0.2.5), both of which exist at Layer L–Q. However, $\hat{F}$'s **physical interpretation** — as the measure of Lie–Jordan mismatch, the "fracture" of $\mathcal{H}_{\text{kin}}$ into sub-Hilbert spaces — requires the Open Expansion machinery of §1. Specifically:

- The **forward kernel inclusion** $\ker(\hat{M}) \subseteq \ker(\hat{F})$ is algebraically provable at Layer Q (if $\omega(\hat{M}) = 0$ then all $\hat{C}_\alpha$ vanish, so all $\Delta_{\alpha\beta}$ vanish, so $\hat{F} = 0$).

- The **reverse kernel inclusion** $\ker(\hat{F}) \subseteq \ker(\hat{M})$ is NOT provable at Layer Q; it requires the Convergence Theorem (§1.6), which shows that R_t drives every state to $\ker(\hat{M}) \cap \ker(\hat{F})$.

The quantity $B_{\Delta}[\rho] := \text{Tr}(\rho \hat{F})$ is a well-defined non-negative number at Layer Q, but it is **not yet named "burden"** — that name requires the physical interpretation (as the cost of the Lie–Jordan mismatch), which is given in §1.3.

Remark 0.3.7 (The F decomposition — deferred). In v2.1, the fracture operator will be decomposed (in §1.2 and §4.4) as:

$$\hat{F} = \hat{F}_{\text{intra}} + \hat{F}_{\text{cross}}$$

where $\hat{F}_{\text{intra}} = \sum_i P_i \hat{F} P_i$ (the intra-sector fracture, capturing constraint violations within each sub-Hilbert space) and $\hat{F}_{\text{cross}} = \sum_{i \neq j} P_i \hat{F} P_j$ (the cross-sector fracture, capturing the cost of maintaining coherence across sectors). This decomposition requires the sector projectors $\{P_i\}$, which come from the fracture of $\mathcal{H}_{\text{kin}}$ (§1.2.1). It is the basis for the v2.1 identification of relational burden as cross-sector burden (§4.4, §5). **[Defined formally in §1.2.4.]**

[GAP: v2.1 — the F decomposition is named here but not formalized. The sector projectors $\{P_i\}$ require the fracture (§1.2), which requires the Open Expansion Principle (§1.1). If the fracture mechanism cannot be made rigorous (which depends on T-2), the F decomposition — and the relational-burden identification — stalls. See Phase C Program Ch 5.]

§0.4 The Two Structural Primitives: Causality and Locality

Definition 0.4.1 (Causality — the Lie Primitive)

Causality is the structural primitive grounded in the Lie algebra $\mathcal{A}_L$. It is the "when"-capacity — the capacity for one event to depend on another, for there to be a directional ordering between relations. Operationally, causality is activated when the Lie bracket $[A, B] \neq 0$ for some observables A, B : this non-commutativity is the algebraic signature of temporal dependency.

The formal definition of the causal order $\prec$ (a strict partial order on zero-preserving events) is deferred to §1.8, after the fracture and the Reconciliation Propagator make the operational meaning of "event" and "causal dependency" precise.

Definition 0.4.2 (Locality — the Jordan Primitive)

Locality is the structural primitive grounded in the Jordan algebra $\mathcal{A}_J$. It is the "where"-capacity — the capacity for observables to co-occur, to be simultaneously measurable, to define spatial sectors. Operationally, locality is activated when the Jordan product $A \circ B \neq 0$ for some observables A, B : this symmetric co-occurrence is the algebraic signature of spatial adjacency.

The formal definition of the localisable sector $\mathcal{A}_{\text{loc}}$ (the set of observables with non-vanishing GNS norm that participate in the emergent geometry) is deferred to §3.1, after the correlation kernel K_ω and the Fubini-Study metric are constructed.

Remark 0.4.3 (The two-primitive picture). The framework's central architectural commitment is the **two-primitive picture**: causality and locality are genuinely independent structural primitives, grounded in the Lie and Jordan structures of the algebra respectively. Neither is inferable from the other (Remark 0.1.3). They are activated jointly by the Open Expansion Principle (§1.1), which drives local Jordan-built sectors to reconcile with the single global causal Lie layer.

This is the replacement for v1.0's " $L \rightarrow Q \rightarrow C \rightarrow Q$ emergence ladder" as the organizing principle of §0. The emergence ladder survives as a layer-labeling scheme (L, Q, C, Q') but is no longer the central architectural commitment — the two-primitive picture is.

Theorem 0.4.4 (Complementarity of the Governing Mechanisms)

[RENAME in v2.1: was "Complementarity / Principle of the Now" in v2.0. The v2.0 name was misleading because the "now" is not defined until §4.2. The theorem text already limited its scope to governing mechanisms; the name now matches.]

The Lie and Jordan structures of $\mathcal{A}$ are complementary governing mechanisms: neither is reducible to the other, and their joint activation (via Open Expansion, §1.1) is necessary for any physical structure to emerge. Specifically:

1. **Independence:** The Lie bracket $[A, B]$ and the Jordan product $A \circ B$ are algebraically independent (Jordan–von Neumann–Wigner, 1934).
2. **Complementarity:** Causality (Lie) provides the "when"; locality (Jordan) provides the "where." Neither alone is sufficient for physical structure — both are needed.
3. **Compatibility:** Associativity of the original product $\cdot$ is the compatibility constraint that allows Lie and Jordan to coexist in a single algebra.

Proof sketch. (1) is the Jordan–von Neumann–Wigner classification theorem. (2) follows from (1): if Lie and Jordan are independent, neither can simulate the other, so both must be present for the full "where + when" structure. (3) is verified by expanding $A(BC) = (AB)C$ in the Lie–Jordan decomposition: the cross-terms cancel iff the product is associative. $\square$

Remark 0.4.5 (What this theorem does NOT claim). Theorem 0.4.4 establishes the complementarity of the **governing mechanisms** (Lie and Jordan). It does **not** define the operational "now" — the joint product of "where" and "when." The "now" is defined in §4.2 (Definition 4.2.1), after both the operational "where" (§3, emergent geometry) and the operational "when" (§4.1, duration) are available. The v2.0 name "Principle of the Now" was therefore premature; the v2.1 name correctly scopes the theorem to the governing mechanisms.

[GAP: v2.1 — none in §0.4. The two-primitive picture is structurally complete at the foundational level. The operational definitions of causality (§1.8) and locality (§3.1) are deferred but their algebraic grounding is established here.]

§0.5 Architectural Summary

What Section 0 Contains

Subsection	Content	Layer
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§0.0	Purpose of the foundation	—
§0.1	Relational algebra, Lie–Jordan decomposition, primitive constraints, $\mathcal{M}$	L
§0.2	Admissibility, constraint ideal, compatibility bracket	L–Q
§0.3	GNS construction, kinematic Hilbert space, $\mathcal{F}$ (algebraic), correlation strength	Q
§0.4	Causality (Lie) and locality (Jordan) as structural co-primitives	L (naming)

What Section 0 Does NOT Contain (Deferred to v2.1 Sections)

Object	Deferred to	Reason for deferral
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Reconciliation Propagator $\$R_t\$$	§1.5	Requires the fracture (§1.2) and burden (§1.3)
Convergence Theorem	§1.6	Requires $\$R_t\$$ (§1.5)
Thin/full $\$A_{\text{phy}}\$$ split	§1.7	Requires the Convergence Theorem (§1.6)
Reduced algebra $\$A_{\text{red}}\$$	§1.4.2	Requires the fracture (§1.2) and Tier 1/Tier 2 split (§1.4.1)
$\$\hat{M}_{\text{red}}\$$ (Tier 2 sub-collective)	§1.4.3	Requires $\$A_{\text{red}}\$$ (§1.4.2)
Dirac bracket	§1.4.5	Requires Tier 1/Tier 2 split (§1.4.1)
Burden $\$B_{\Delta}\$$ (as physical quantity)	§1.3	Requires the fracture (§1.2) for physical interpretation
Reconciliation Principle	§2.1	Requires burden (§1.3) and events (§1.8)
Correlation kernel $\$K_{\omega}\$$ / $\$K_{\omega}^J\$$	§3.1	Requires $\$A_{\text{loc}}\$$ (§3.1, which requires $\$A_{\text{phy}}^{\text{thin}}\$$ from §1.7)
Fubini-Study metric $\$d_{\text{FS}}\$$	§3.2	Requires $\$K_{\omega}\$$ (§3.1)
F decomposition ($\$\hat{F}_{\text{intra}}\$ + \$\hat{F}_{\text{cross}}\$)$	§1.2.4	Requires sector projectors from the fracture (§1.2)
Adjacency structure on $\$\hat{C}_{\alpha}\$$	§1.1	Requires Jordan co-occurrence (activated by Open Expansion)
Causal order $\$\prec\$$	§1.8	Requires zero-preserving events (§1.8.1)
The "now"	§4.2	Requires space (§3) and duration (§4.1)

Acyclicity Test

Question: Does Section 0 define every object before using it?

Verification:

- §0.1 uses only the algebra A (primitive). ✓
- §0.2 uses A , M (from §0.1). ✓
- §0.3 uses A , ω , M , $\Delta_{\alpha\beta}$ (from §0.1–0.2). ✓
- §0.4 uses A_L , A_J (from §0.1). ✓
- §0.5 (this subsection) references deferred objects but does not use them. ✓

Forward references: None that create circularity. The objects named in §0.3.7 (F decomposition) and §0.2.7 (adjacency structure) are named as forward pointers to §1, not used in any definition or proof within §0.

Verdict: Section 0 is acyclic. The architecture holds at the foundational layer.

Inline Gap Summary for §0

Gap ID	Location	Description	Severity	Blocks
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GAP-0.1.5	§0.1 Remark 0.1.5	Representation-theoretic claim " A_J supports exactly 3 independent symmetric bilinear invariants" not proven	FATAL	D=3 closure (§3.6)
GAP-0.1.7	§0.1 Remark 0.1.7	Adjacency structure on constraint index set I not yet defined	FATAL	T-1 (§1), T-2 (§1.6)
GAP-0.2.7	§0.2 Remark 0.2.7	Jordan component $\hat{C}_\alpha \circ \hat{C}_\beta$ named but not given operational meaning	MATERIAL	Adjacency structure (§1.1)
GAP-0.3.4	§0.3 Remark 0.3.4	K_ω^J vs K_ω : which is the correct kernel?	MATERIAL	§3.1, D=3 (§3.6)
GAP-0.3.7	§0.3 Remark 0.3.7	F decomposition named but not formalized	MATERIAL	Relational burden (§4.4)

Total new gaps surfaced in §0: 5 (2 fatal, 3 material). All are pre-existing gaps from the Phase C program, now flagged at their point of origin. No new gaps introduced by the v2.1 rewrite of §0.

Section 0 is CLOSED. Section 1 (Open Expansion, Fracture, and the Reconciliation Propagator) can be merged against this stable foundation.